

Day 3 Task Report: Exploratory Data Analysis (EDA)

Date: June 11, 2026

Project Role: Data Science Intern

Dataset: Iris Dataset

Tools Used: Python (Pandas, Matplotlib, Seaborn)

1. Executive Summary

The primary objective of Day 3 was to conduct a comprehensive **Exploratory Data Analysis (EDA)** on the classic Iris dataset. By leveraging statistical summaries and data visualization techniques, the goal was to uncover underlying patterns, detect relationships between features, and prepare the dataset for future predictive modeling.

2. Methodology & Implementation

Phase 1: Summary Statistics

The dataset consists of 150 records across 3 species of Iris (*Setosa*, *Versicolor*, *Virginica*), with 4 features: sepal length, sepal width, petal length, and petal width.

- **Data Integrity Check:** Handled using `df.info()` and `df.isnull().sum()`. The dataset contains zero missing values.
- **Descriptive Statistics:** Generated using `df.describe()` to evaluate the mean, median (50%), standard deviation, and quartiles for each feature.

Phase 2: Identifying Patterns & Distributions

To understand how the data is distributed and how the species separate visually, the following plots were generated:

- **Histograms & KDE Plots:** Used to analyze the distribution of individual features. Petal length and width showed clear bimodal distributions, hinting at distinct species separation.
- **Pairplots (sns.pairplot):** Plotted a matrix of scatter plots across all features, color-coded by Iris species.
 - *Observation:* *Iris-setosa* forms a completely distinct cluster from the other two species across almost all feature combinations.

Phase 3: Correlation Analysis

A correlation matrix was calculated using Pearson's correlation coefficient to measure the linear relationships between numeric features.

- **Heatmap Implementation:** Visualized using `sns.heatmap()` with an annotated color scale.

3. Key Findings & Insights

- **High Feature Correlation:** Petal length and petal width have an exceptionally strong positive correlation (~ 0.96). If a plant has a longer petal, it almost certainly has a wider petal.
- **Sepal/Petal Relationship:** Petal length is also highly positively correlated with sepal length (~ 0.87). However, sepal width shows a weak negative correlation with sepal length.
- **Species Separability:** * *Setosa* can be linearly separated from *Versicolor* and *Virginica* using only petal measurements.
 - *Versicolor* and *Virginica* have slight overlap but remain largely distinguishable based on feature thresholds.

4. Final Deliverables Status

- **EDA Report:** Completed (This document).
- **Visual Graphs:** Saved locally as .png files (pairplot.png, correlation_heatmap.png, feature_distributions.png).
- **Outcome:** Successfully achieved. Developed a solid understanding of structural patterns, data distributions, and feature dependencies within the dataset.